

CHE 473K

Process Design and Operations

Spring 2026 (16630)

Course Description: The primary aim of this capstone course is to apply the fundamentals of chemical engineering acquired from all your previous courses to the design and operation of chemical processes. Students will learn how to design a chemical process, to estimate the fixed and operational costs of the process, and to evaluate the overall economics of a process using the principles of the time value of money and discounted cash flow analysis. Students will also learn about the health, safety, and environmental aspects of their profession.

Prerequisites: ChE 363 (Separation Processes and Mass Transfer), and ChE 372 (Chemical Reactor Analysis and Design).

Lecture Time: M/W/F noon – 1:00pm in CPE 2.216

Recitation Time: F 2:00pm – 3:50pm in CPE 2.204

Instructor: David T. Dalle Molle (512-471-7722 | dtd@che.utexas.edu)

Office Hours: Monday & Friday 11-12am (in CPE 3.458) or by appointment.

TA office hours may be moved after the first 6-8 weeks as the focus shifts to the Design Project.

Recitation Session: The recitation sessions will cover:

- 1) Tutorials on the use of ASPEN Plus (Computer based process design)
- 2) Homework review / solution presentation
- 3) Supplemental Office hours, Design Project Team Building and help sessions

This time slot will also be used for some Exams and Design Project Presentations.

Text: None required. There will be some recommended resources posted in Canvas.

Grading basis: Homework/Tutorials/Quizzes:	15 %
Exam No. 1	30 % (Thu Feb 26 th 7-9pm)
Exam No. 2	15 % (Wed Apr 1 st 7-9pm)
Design Project - Presentations	10 % (Final Presentations Apr 24 11a-6p)
Design Project - Reports	30 % (Final Report Due 5pm Apr 27)

Note that grades will be curved as necessary and letter grades assigned based on natural breaks in the distribution. Plus/Minus grading will be used in this course.

General Course Outline. The course will cover the following topics:

- 1) Process design
- 2) Fixed Capital Estimation
- 3) Computer simulation
- 4) Process analysis
- 5) Process economics
- 6) Special topics: public health, process safety, environmental compliance, ethics, and sustainability
- 7) Comprehensive design project

Note: Information posted in the Announcements section in Canvas ([Sp26 - PROCESS DESIGN AND OPERATIONS \(16630\)](#)) supersedes that in this syllabus. It is your obligation to read and note any changes in the Canvas Announcements.

Course Objectives

Develop professional skills in the development, design, and operation of chemical processes involving a variety of unit operations (absorption/stripping, distillation, extraction, adsorption, crystallization, chemical conversions, etc.) and a variety of industrial sectors (petroleum, chemical, power production, microelectronics, biomedical, environmental, etc.).

Provide exposure to process equipment design and operations through class lectures and individual study during team project execution.

Provide training and experiences in developing communication skills (reading, writing, oral presentations, listening) through formal course training, team project execution, team project presentations, team project reports, and class exercises.

Provide training and experiences in sound problem solving approaches [define the problem and what is known, what additional information is needed, how do we obtain/measure the needed information, what techniques can be used to solve the problem, what is the best approach to obtain a good (timely, adequate accuracy, and cost effective) solution, and does this solution make sense/what is the best way to verify it.]

Provide adequate training and experiences in the use of engineering tools such as computer process simulation and economic models.

Provide learning environment that stimulates students' curiosity and interest in solving challenging engineering problems.

Provide learning environment that encourages students to "learn on their own" and feel responsible for their own career development.

Provide a learning environment that encourages students to conduct their professional activities in a fashion consistent with the engineering code of ethics.

Knowledge, Abilities, and Skills Students Should Have Entering This Course:

1. Basic Math, Chemistry, and Physics
2. Set-up and solve macroscopic mass balances (ChE317 and ChE354)
3. Set-up and solve macroscopic energy balances (ChE317 and ChE354)
4. Ability to size piping networks, valves, pumps, compressors, and flow meters for simple and branched flow systems (ChE354).
5. Ability to determine heat transfer coefficients and size heat exchangers (ChE354).
6. Ability to calculate physical and chemical equilibrium for ideal and non-ideal systems (ChE322, ChE363, ChE372)
7. Ability to size various separations systems, specifically distillation, absorption, adsorption extraction, and membrane separation processes (ChE363).
8. Ability to analyze various reaction systems and size chemical reactors for specific reaction systems (ChE372).
9. Basic computer skills (ChE210) and numerical methods (ChE348).
10. Can use process simulation models to analyze the performance of a separations' unit operations, such as distillation (ChE363).
11. Write reports (ChE333T, ChE333K) and make oral presentations (ChE333T).

Knowledge, Abilities, and Skills Students Should Gain From This Course:

1. Define the approach and solve process design problems.
2. Develop confidence in solving open-ended problems.
3. Obtain appropriate data for solving problems.
4. Perform cost estimations, profitability analyses, and alternative investment evaluations.
5. Select appropriate process equipment and materials of construction.
6. Effectively analyze and troubleshoot results obtained from process simulation models.
7. Develop project management plans.
8. Understand typical process development steps.
9. Prepare effective project reports.
10. Give effective oral presentations.
11. Develop an appreciation of process designs in several industries.
12. Develop an appreciation for process safety, product liability, and environmental control issues.
13. Develop an appreciation for the importance of engineering professionalism.

Student Outcomes

CHE 473K is a capstone course, and it will help provide Student Outcomes 2, 3, 4, 5 and 7 of the Accreditation Board for Engineering and Technology (ABET):

2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. An ability to communicate effectively with a range of audiences

4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies

Course Format & Learning Activities:

- The reference material and supplemental notes provide excellent coverage of the application of process design methodologies. These readings are intended to provide you with some theoretical coverage of the field but will typically emphasize the practical application of design principles.
- Lectures will complement the reference material and class handouts. In general, lectures will be interactive, combining in-class discussions with small group problem solving exercises, chalk board problem analyses, etc. Open-ended questions, similar to problems you may encounter in your later engineering practice, will provide you with opportunities to enhance your problem formulation and problem-solving skills. Videos on how certain pieces of equipment function will provide you with the ability to identify pieces of equipment and will also help you to link the theory with the physical.
- Recitation sections will be overseen by the TA and/or the Professor and will focus on the solution of additional relevant problems. In some cases, we will cover difficult homework problems and go through exam solutions. In all cases, an emphasis will be placed on how a problem is approached and why a particular approach is selected.
- Regular homework assignments on the technical material will be given to provide practice applying the concepts covered in lecture. Group discussion of the homework (but not copying) is encouraged. However, it is important for exams that each student know how to independently work the assigned problems on his/her own.
- Two exams will test both retention of concepts and facts, and the ability to apply problem-solving skills. Material tested will be extracted from the readings, homework, lectures, guest speakers, projects, and recitations.

Performance Feedback:

Feedback on your performance throughout the semester is a key component to the learning process. In addition to feedback on homework and exams through solution sets, comments and grades, you will receive immediate input through in-class problem-solving and group activities. Feedback will also be actively encouraged through visits with the instructor or the TA in office hours or by appointment. However, you should make a genuine attempt at all problems on your own prior to seeking additional help from the instructor or TA.

Course Policies:

Attendance – Attendance at lectures and recitation sessions, although not mandatory, is important for your mastery of the subject matter.

Homework & Projects – Homework, Tutorials, and project related assignments must be turned in at the due date/time shown in Canvas.

Homework must follow the guidelines from ChE 317: engineering paper should be used, problem statement must be defined, sources of information (Appendices, etc.) must be identified, and an arrow must point to a single boxed solution. Illegible papers or those not adhering to the criteria established in ChE 317 will not be graded. Solutions to all problems must include adequate steps, and explanations where necessary, for us to understand how you arrived at an answer. Just an equation and a final answer will not be acceptable.

Grading Policies - Assignments will be graded considering the following factors:

- Participation (making any honest attempt)
- Accurately defining the problem (drawing a correct schematic/interpretation of the system)
- Setting up the problem (making accurate assumptions, defining the correct equations, adequately explaining all steps required to reach a solution, justifying your approach)
- Correctness and completeness (making accurate calculations, using correct conversions, using appropriate physical properties, identifying sources of information, accurately interpreting data from tables, carrying solution to completion, using correct significant figures)
- Neatness, organization (logical flow), and boxing final solutions

Problems should be worked in units that are consistent with the units given in the problem statement. The final solution must be reported in units consistent with those given in the problem statement. For example, if the problem statement gives metric units, do not report your solution in English units. Note: Conceptual errors are weighted more heavily than simple mathematical (calculation) errors.

Late Assignments – No late assignments will be accepted without permission from the instructor. Requests must be made in advance of the assignment deadline.

Regrading: If you feel an exam problems or homework assignment was not correctly or fairly graded, or that it was simply not totaled correctly, you may submit a regrade request that must include a specific explanation for the request relative to the published solution/rubric. Only math errors and oversights will be considered valid reasons for dispute. If your work is not clear or the specific question involves subjectivity, then there is no justification for re-grading.

All regrade requests must be submitted through Gradescope within 48 hours of when the graded document is available for review to the class. Failure to promptly review your graded exams and homework assignments soon after they are published in Gradescope does not entitle you to any

additional time to make a regrade request. Note that it is possible that regrading could result in a lower grade should other issues come to light in the review.

Exams – All exams will be closed book and closed notes. No electronic versions of any books or reference material will be allowed during the exam.

As mentioned above, requests for re-grading of exams must be submitted via Gradescope within 48 hours after the graded exam has been returned. If re-grading is desired, then the entire exam is subject to re-grading. As with Homework assignments, if your work is not clear or the specific question involves subjectivity, then there is no justification for re-grading.

Missed Exam – No make-up exams will be given without a written excuse for the absence provided at least one week in advance of the exam.

Expectations of Conduct:

We are privileged to participate in the pursuit of knowledge and truth in higher education, and students and instructors are expected to maintain an environment of respect for the course of study and one another at all times. Our classroom is a safe space for people diverse in traits and ideology to exchange ideas and grow in experience and knowledge. All students are welcome in our course, and we will respect our differences including those in gender, race, ethnicity, nationality, religion, sexual orientation, gender identity, age, culture, experience, and socio-economic background. No form of excessive teasing, discrimination, or bullying shall be tolerated at any time. Concerns about classroom environment should be addressed immediately to the instructor.

Academic Integrity

Students who violate University rules on scholastic dishonesty (including cheating on tests, plagiarism on written assignments, collaboration on assignments that are specifically intended to be non-collaborative, etc.) are subject to disciplinary penalties, including the possibility of failure in the course and/or dismissal from the University. Since such dishonesty harms the individual, all students, and the integrity of the University, policies on scholastic dishonesty will be strictly enforced. It is each and every student's responsibility to be familiar with the University rules and guidelines on academic integrity and scholastic dishonesty, which are posted at [Office of the Dean of Students - Student Conduct and Academic Integrity - Standards of Conduct](#).

Expectations of Academic Integrity

We must respect one another's ideas by giving credit where it is due, avoiding all forms of plagiarism and cheating. Any item submitted by you and that bears your name is presumed to be your own original work that has not previously been submitted for credit in another course.

You may use words or ideas of other individuals from publications, web sites, or other sources, but **only with proper attribution**. "Proper attribution" means that you have fully identified the original source and extent of your use of the words or ideas of others that you reproduce in your work for this course, usually in the form of an endnote.

Students should be aware that all required writing assignments may be submitted to a plagiarism detection tool such as Turnitin.com. Turnitin is a software resource intended to address

plagiarism and improper citation. The software works by cross-referencing submitted materials with an archived database of journals, essays, newspaper articles, books, and other published work. In addition, other methods may be used to determine the originality of the paper. This software is not intended to replace or substitute for the faculty member's judgement regarding detection of plagiarism.

Use of Artificial Intelligence (AI) Tools – The creation of artificial intelligence tools for widespread use is an exciting innovation. These tools have both appropriate and inappropriate uses in classwork. The use of artificial intelligence tools (such as ChatGPT) in this class is strictly prohibited. This includes using AI to generate ideas, outline an approach, answer questions, solve problems, or create original language. All work in this course must be your own or created in group work, where allowed.

If you are not clear about the expectations for completing an assignment, be sure to seek clarification from the instructor beforehand.

Finally, you should keep in mind that as a member of the campus community, you are expected to demonstrate integrity in all of your academic endeavors and will be evaluated on your own merits. So be proud of your academic accomplishments and help to protect and promote academic integrity at the University of Texas.

Title IX

Title IX, a segment of the Education Amendments of 1972, makes it clear that violence, harassment, or discrimination based on sex and gender are Civil Rights offenses subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories such as race, national origin, sexuality, disability, etc. If you or someone you know has been harassed or assaulted, please don't hesitate to speak to anyone on our teaching team. You can also find useful resources at [Support & Resources | Title IX Office](#).

Sharing of Course Materials is Prohibited

No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class without my explicit, my written permission. Unauthorized sharing of materials may facilitate cheating. The University is aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to [Student Conduct and Academic Integrity](#) in the Office of the Dean of Students. These reports can result in initiation of the student conduct process and include charge(s) for academic misconduct, potentially resulting in sanctions, including a grade impact.

Class and Recitation Recordings

Class and recitation recordings are reserved only for students in this class for educational purposes and are protected under FERPA. The recordings should not be shared outside the class in any form. Violation of this restriction by a student could lead to Student Misconduct proceedings.

Accommodation of Special Situations and Needs

If you need accommodations related to physical, psychological, or learning abilities, please speak to the instructor after class or during office hours.

If you must miss class because of religious observation, holy day, or off-site interview please speak to the instructor after class or during office hours, at least one week prior to the absence, in order to make arrangements to submit work early. It is your responsibility to review materials outside of class on your own to make up for class time missed.

Academic Accommodations: The University of Texas at Austin provides upon request appropriate academic accommodations for qualified students with disabilities. For more information, contact the [Division of Diversity and Community Engagement, Accommodations and Services](#) at access@austin.utexas.edu or 512-471-6259 (audio).

Classroom Safety and COVID-19: To help preserve our in person learning environment, please utilize the resources available at the [Healthy Horns website](#).